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VISION

----- Active space -----

Vision is a special sense.

Organisation of vision : 3 levels

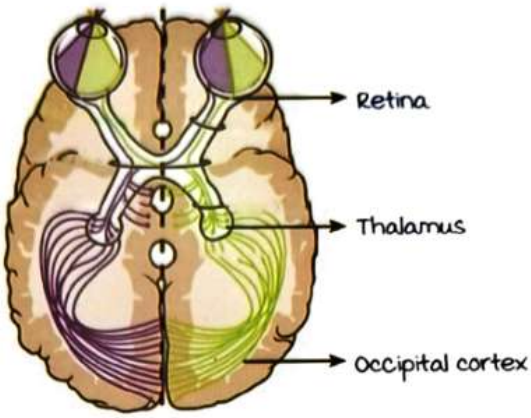
Retina

↓

Thalamus

↓

Occipital Visual cortex



Organisation of vision

Note :

General senses : Have 4 levels of organisation.
(Includes spinal cord).

Retina

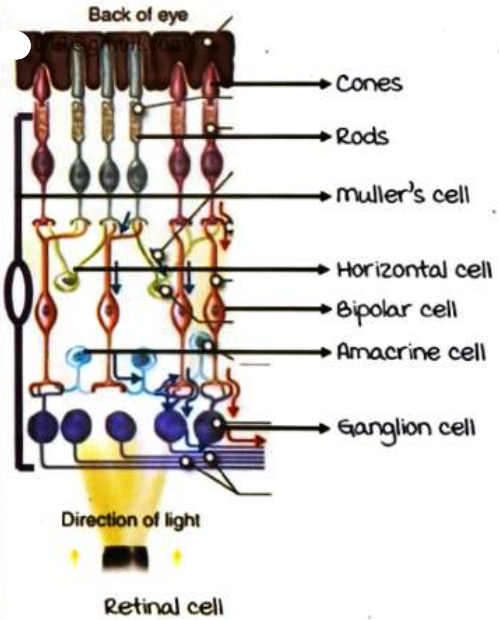
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7 Types of cells :

- 1st – 6th : Play a role in vision.
- 7th : Supporting cell.

RETINAL CELLS :

Retinal cell	Features
1. Rods	Rod shaped
2. Cones	Cone shaped
3. Bipolar cell	Has 2 poles → One to receive information. → One to conduct information.
4. Ganglion cells (GC)	Only retinal cell to : - Have an output. Axons → Optic nerve → Thalamus. - Generate action potential (For impulses to travel long distances).
5. Horizontal cell	Connects cones & rods
6. Amacrine cell	Connects bipolar cell to ganglion cells
7. Muller's cell (AKA retinal glial cell)	• AKA retinal glial cell. • Supportive function.



Retinal cell

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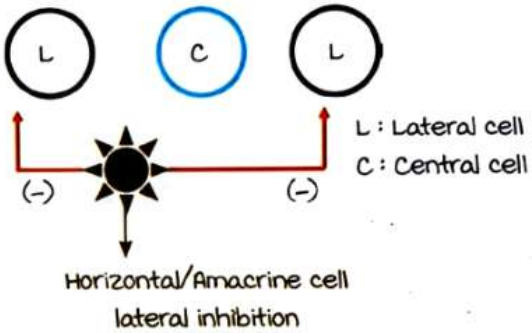
----- Active space -----

Photoreceptors :

	Rods	Cones
Sensitive to	Dim lights (Night vision)	Colour vision (Day vision)
Number	100 million	5 million
Pigment	Rhodopsin/visual purple	Iodopsin
Abundant in	Peripheral retina	• Central retina. • Foveola : Only cones (Rod free zone).

Lateral inhibition :

Horizontal/Amacrine cell
↓
Inhibits lateral cells
↓
Enhanced output from central cell
↓
↑ contrast & sharpness



Melanopsin ganglion cells (MGC) :

- MGC → Directly stimulated by light.
- Pigment : Melanopsin.
- Contains PACAP : Pituitary adenylate cyclase activating polypeptide → For signalling.

Projections of MGCs

- Hypothalamus : Retinal hypothalamic tract (RHT) → Suprachiasmatic nucleus : Regulates 24hr circadian rhythm
- Pretectal nucleus → Edinger westphal nucleus : Regulates pupillary light reflex

PHOTOTRANSDUCTION :

Changes in the retina when light strikes it.

Rods :

- Contain rhodopsin → Opsin + Vitamin A.
- Forms of Vit A
  - 11-cis retinal (Dark state)
  - All-trans retinal (Light state)

Feedback

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In dark state :

RMP of rods = -60mV (Passive K^+ diffusion).

Funny current channel remains open

↓

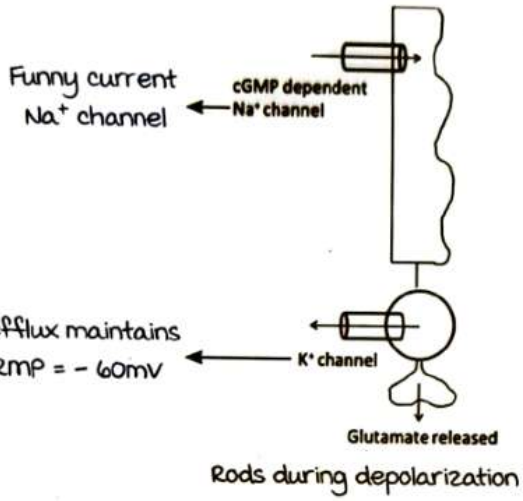
Allows Na^+ influx

↓

Depolarization

↓

Glutamate release (Neurotransmitter)



Rods during depolarization

In light state :

11 cis retinal

↓ light

All trans retinal

↓ (+)

Transducin

↓ (+)

Phosphodiesterase

↓

Degrades cGMP

↓

Closure of funny Na^+ channel

K^+ channels remain open

↓

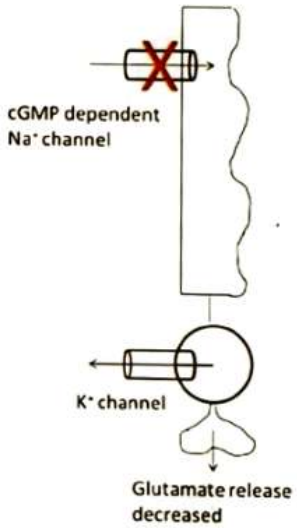
K^+ efflux

↓

Hyperpolarisation

↓

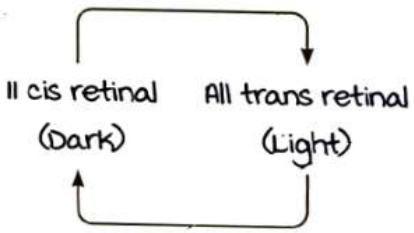
↓ glutamate release



Hyperpolarisation

Wald's visual cycle :

Constant cycling between 2 forms of Vitamin A during day & night.



Funny current channels/hyperpolarisation cyclic nucleotide (HCN) channels :

- Na^+ channels.
- Location & function :
  - Retina : Vision.
  - SA node : Generation of HR.

Opened by cGMP.

Amacrine cells :

Purely depolarizing cells of retina (No hyperpolarisation).

Role in lateral inhibition.

Neurotransmitter : Acetylcholine.

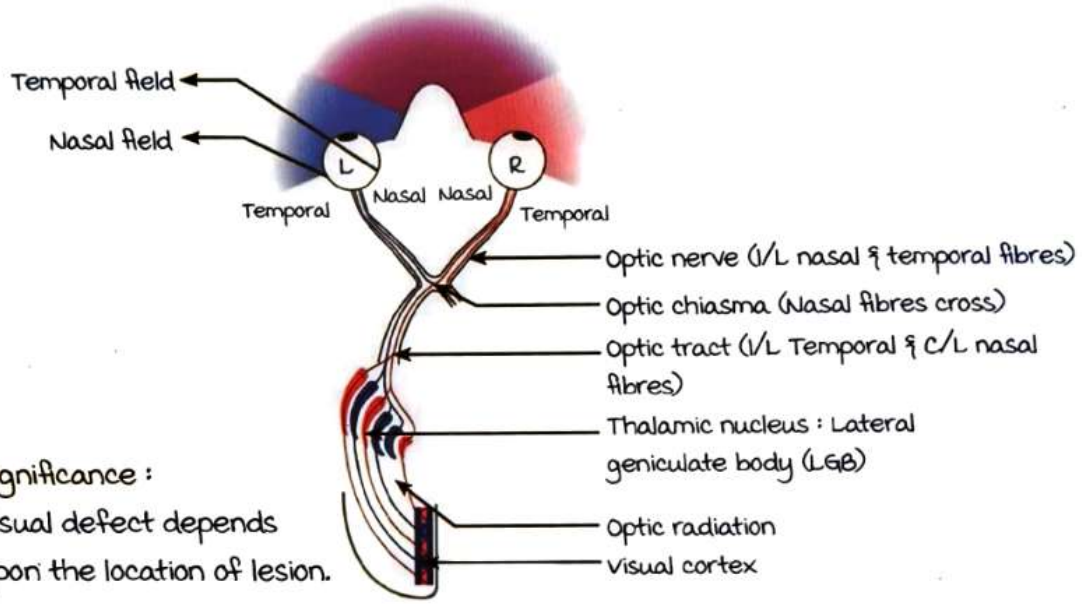
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Visual Pathway

00:33:00



Significance :
Visual defect depends upon the location of lesion.

Thalamus

00:38:00

LATERAL GENICULATE BODY

Has 6 layers.

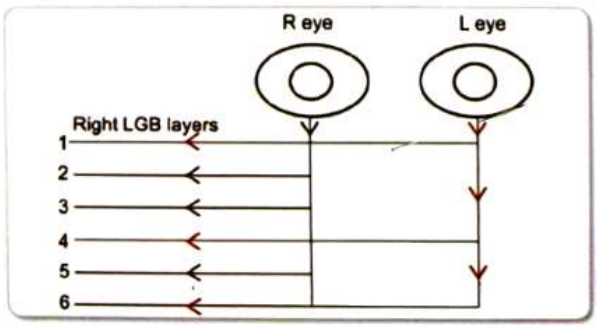
	Magnocellular pathway (M cells/big)	Parvocellular pathway (P cells/small)	Koniocellular pathway (K cells)
Origin	1, 2 layers	3, 4, 5, 6 layers	Between P&M cells.
Termination	Layer 4 of visual cortex	Layer 4 of visual cortex	-
Function	• Extraocular movements. • Detection of flickering.	• Colour vision. • Finer details in the visual field.	Colour vision : Blue

INPUTS TO LGB :

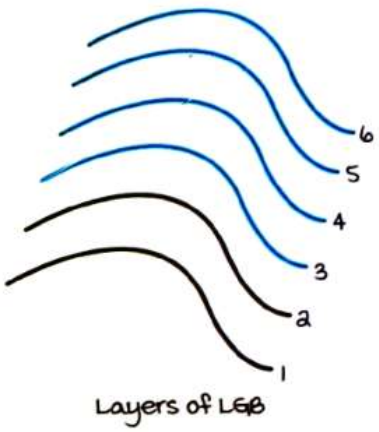
I/L LGB receives input from I/L & C/L eye.

Layers 1, 4, 6 → Receive C/L fibres.

Layers 2, 3, 5 → Receive I/L fibres.



Right LGB fibres



Layers of LGB

Feedback

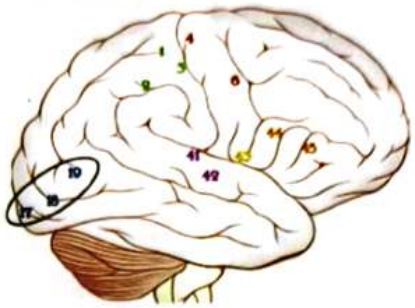
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Visual Cortex

00:46:21 ----- Active space -----

LOCATION

- Calcrine cortex (In occipital lobe) : Brodman's area 17, 18, 19.
- Heavily myelinated.
- Stripes / striae appearance : Striae of Gennasi. (AKA striate cortex).



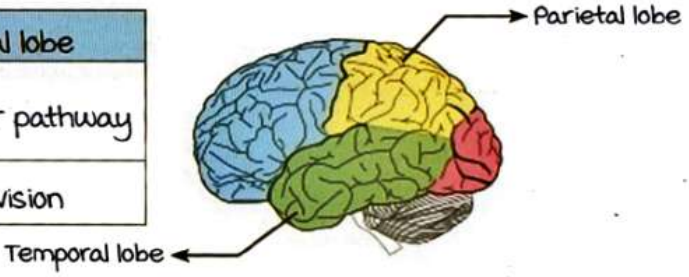
Visual cortex

CELLS :

	Simple cell	Complex cell	Blobs
Function	• Higher order processing. • Feature detectors : Detect minute changes.		• Colour vision. • Expresses cytochrome oxidase.

EXTRA STRIATE CORTICES :

	Parietal lobe	Temporal lobe
Projections from	Magnocellular pathway	Parvocellular pathway
Function	Eyeball movements	Colour vision



Extrastriate cortex

Inferior temporal cortex :

Function : Face recognition by sight.

Lesion : Prosopagnosia → Loss of recognition of faces.

Colour Vision

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PATHWAY

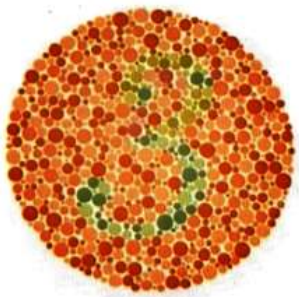
Retina : Cones

↓

Thalamus : Parvocellular pathway

↓

Visual cortex : Blobs



Ishihara chart

CONES :

Primary colours : Red, blue, green.

Types of cones :

	L cone	M cone	S cone
Wavelength detection	Long	Medium	Small
Colour	Red	Green	Blue

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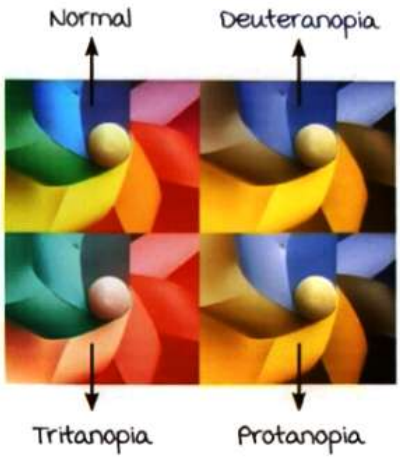


----- Active space -----

COLOUR BLINDNESS :

- X-linked recessive disorders.
- Affects males > females (Carriers).
- m/c : Red green defect.

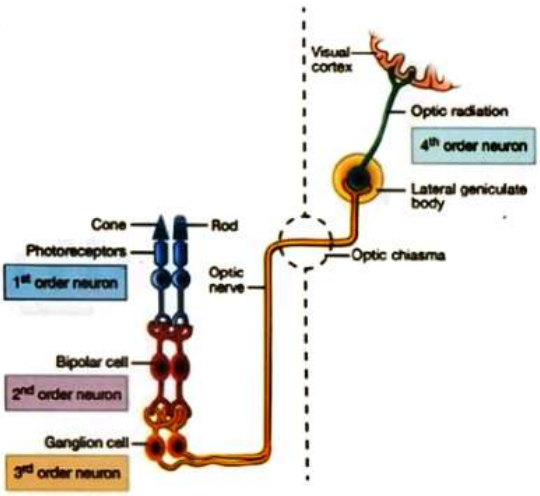
Colour defect	Name
Red	Protanopia
Green	Deuteranopia
Blue	Tritanopia



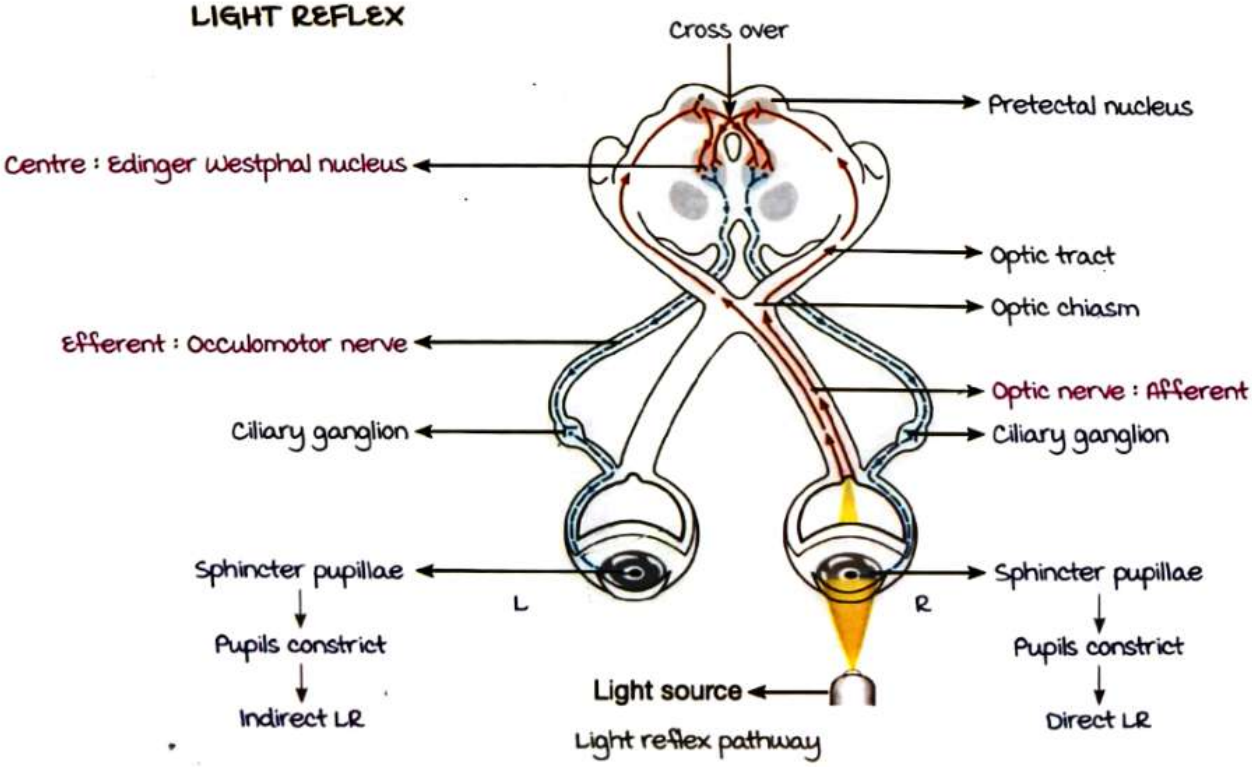
Order of neurons & light reflex

01:01:10

ORDER OF NEURONS



LIGHT REFLEX



Feedback